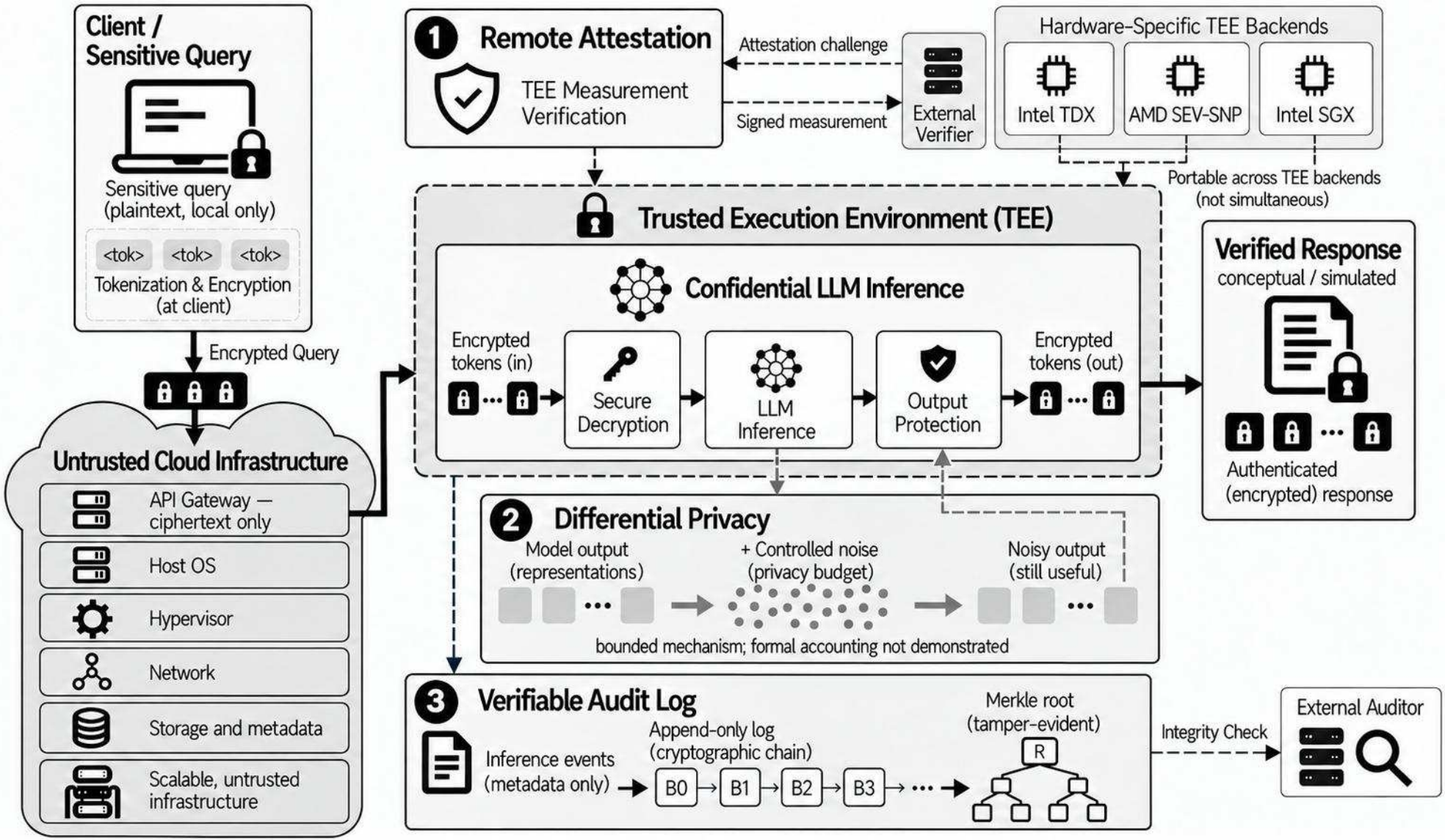

Client / Sensitive Query
Sensitive query (plaintext, local only)
<tok> <tok> <tok>
Tokenization & Encryption (at client)
Encrypted Query
Untrusted Cloud Infrastructure
API Gateway — ciphertext only
Host OS
Hypervisor
Network
Storage and metadata
Scalable, untrusted infrastructure
1 Remote Attestation
TEE Measurement Verification
Attestation challenge
Signed measurement
External Verifier
Hardware-Specific TEE Backends
Intel TDX
AMD SEV-SNP
Intel SGX
Portable across TEE backends (not simultaneous)
Trusted Execution Environment (TEE)
Confidential LLM Inference
Encrypted tokens (in)
Secure Decryption
LLM Inference
Output Protection
Encrypted tokens (out)
Verified Response
conceptual / simulated
Authenticated (encrypted) response
2 Differential Privacy
Model output (representations)
+ Controlled noise (privacy budget)
Noisy output (still useful)
bounded mechanism; formal accounting not demonstrated
3 Verifiable Audit Log
Inference events (metadata only)
Append-only log (cryptographic chain)
B0
B1
B2
B3
Merkle root (tamper-evident)
R
Integrity Check
External Auditor